S-Block Elements

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S-Block Elements

Introduction:

Elements of first(alkali metals) and second (alkaline earth metals) group of representative elements are called s-block elements. All the elements of these groups are metals and have their outer electrons in s-orbital (reason why they are called s-block). In this chapter we will discuss general characteristics of these elements and their compounds. We will also discuss production and uses of different compounds like Portland cement, biological significance of Na, K, Mg and Ca.

GROUP 1 ELEMENTS: ALKALI METALS

First group elements(Li, Na, K, Rb, Cs, Fr) are called alkali metals. They show regular trends in their physical and chemical properties as a function of atomic number, some of such properties are explained below.

Alkali metals

(a) General electronic configuration = ns^1

Elements

Li, Na, K, Rb, Cs } metal

Fr – Radioactive metal

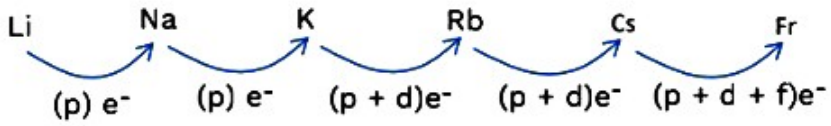
Having one valence electron, lightly bonded to the nucleus, they have high electropositivity. Their behavior of losing an electron easily makes them difficult to be found in free state. They form monovalent ions (M+) on oxidation.

Element	Symbol	Electronic configuration
Lithium	Li	$1s^2s^1$
Sodium	Na	$1s^22s^22p^63s^1$
Potassium	K	$1s^22s^22p^63s^23s^23p^64s^1$
Rubidium	Rb	$1s^22s^22p^63s^23p^63d^{10}4s^24p^65s^1$
Caesium	Cs	$1s^22s^22p^63s^23p^63d^{10}4s^24p^64d^{10}5s^65p^66s^1$ or [Xe] $6s^1$
Francium	Fr	[Rn] $7s^1$

(b) Atomic size

Li
Na
K
Rb
Cs
Fr
} increases ↓

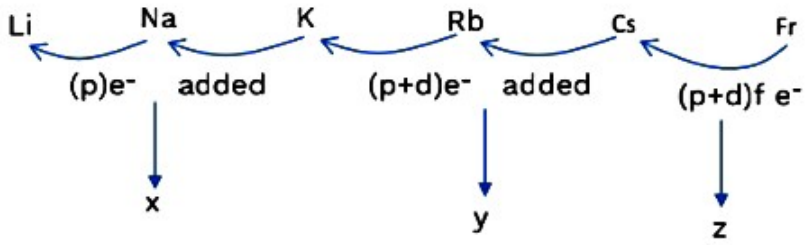
Effective nuclear (Zeff) charge remain constant on moving top to bottom, and n (number of shell) increases from top to bottom.



We also observe that,

On moving from **Li to Na** and from **Na to K**	-electron are added in p -subshell
From **K to Rb** and from **Rb to Cs**	Electrons are added in p subshell and d subshell
From **Cs to Fr**	Electrons are added in p,d,and f subshell.

Hence, number of **protons and electrons increases in same number**.
Due to number of protons increasing in it, it has the same effect.
But due to electrons being added in different subshell-different shielding effect occurs.
As we know, effect of shielding: s > p> d > f
{d and f subshell show negligible shielding}
⇒ Effective nuclear charge has low impact on outer surface of an atom due to poor shielding effect.
Let x= difference in size due to addition of electron in p subshell
y= difference in size due to addition of electron in (p +d) subshell
z= difference in size due to addition of electron in(p+d+f) subshell



Hence x > y > z
Due to negligible shielding of d and f electrons.
Which means size increase but relatively increase is not too high
[due to addition of electron in d and f subshell].

(c) Atomic and Ionic Radius

Atomic:

Increase down the group, because value of principal quantum number increases.

Li > Na > K > Rb > Cs.

Ionic:

One electron is present in valence shell.

$M + e^- \rightarrow M^+$ (nearest noble gas configuration achieved)

Size: $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+ < Fr^+$

(d) Ionization Enthalpy:

As we know, Ionisation. Energy $\propto \frac{\text{Charge}}{\text{size}}$

Size comparatively increases at higher rate as compared to effective nuclear charge on moving from top is bottom along group .
Hence for ionisation potential decreases on moving down the group.

Li > Na > K > Rb > Cs

(e) Hydration Enthalpy and Size:

Hydration energy is the amount of energy released when 1 mole of ion 1 dissolves in water.

(1) Hydration Enthalpy:

Size: $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$ (same charge)

[smallest]

Charge density = $\frac{\text{Charge}}{\text{Radius}}$

Since Li^+ has highest charge density.
More attraction with water, highest energy released order of hydration energy :
$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

(2) Hydration size:

$$\text{Size} \propto \frac{1}{\text{Hydrated size (ion + water)}}$$

Points to remember

Water attracted by ions :
Li^+ (high charge density)
Surrounded by more water molecule (hydrated size)
Example $LiCl \cdot 2H_2O$

Hydrated size : ion collectively with water molecule.
Order : $Li^+(ag) > Na^+(ag) > K^+(ag) > Rb^+(ag) > Cs^+(ag)$

(f) Physical Properties

(1) All the alkali metals are soft, silvery white, and light metals.

(2) Melting point :
A weak metallic bonding, due to presence of single valence electron, leads to low melting point.

$$\text{Radius} = \frac{\text{Metallic bond length}}{2}$$

Points to remember

If radius High – Metallic Bond length high
That means metallic bond weakens and so does the energy required to break metallic bond.
Result
Melting point Li > Na > K > Rb > Cs

(3) Density :

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Mass increases at higher rate in comparison to volume in general, density increases on moving from top to bottom.

Exception :

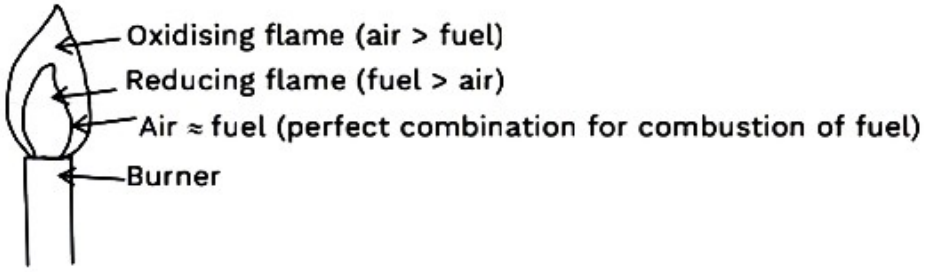
$$\text{Increase in mass}\left\{\begin{array}{l}{}_{3}Li = 1s^2, 2s^1 \\ {}_{11}Na = 1s^2, 2s^2, 2p^6, 3s^1 \end{array}\right\}\{\text{All subshell filled}\}$$
$${}_{18}K = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^0, 4s^1 \text{ (In complete subshell)}$$

Hence order of density → Li **< K < Na** < Rb < Cs
Order reversed in **K and Na** , due to decrease in volume { incomplete subshell} with respect to same increase in mass.

(4) Alkali metals are highly reactive. So, they are kept in kerosene oil except lithium.
Lithium is kept in paraffin wax.
Reason – Due to low density lithium floats on kerosene.
Hence pressed inside wax.

Points to remember

In Bunsen burner, fuel burns in presence of air



- Platinum or nichrome wire made to loop.
- (NaCl + HCl) paste is put on loop and placed on oxidizing flame.
- Na^+ electrons get excited and vibration occurs, due to absorption of energy. Due to which there occurs flame coloration.

(5) Photo electric effect :

(i) Metals with low ionization energy show high photo electric effect
That is {K ,Rb, Cs}.

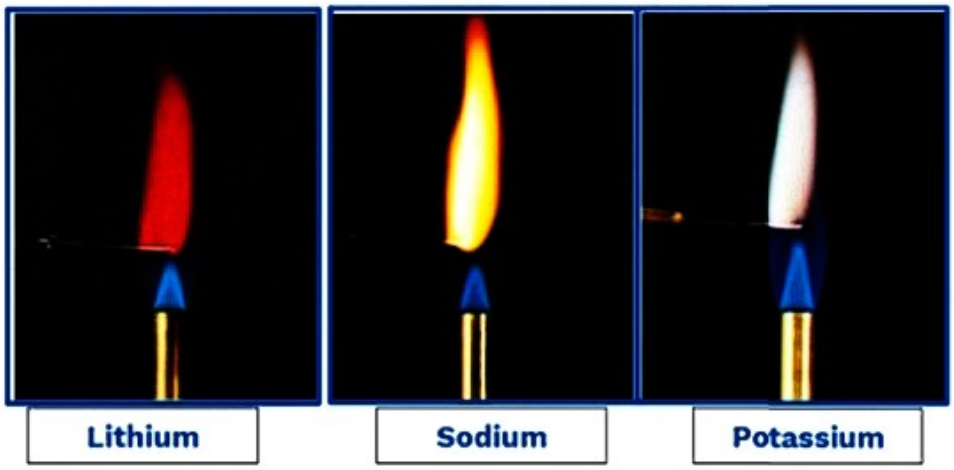
(ii) Only these elements show photo electric effect to due to lowest ionisation potential in periodic table.

(6) Flame Test :

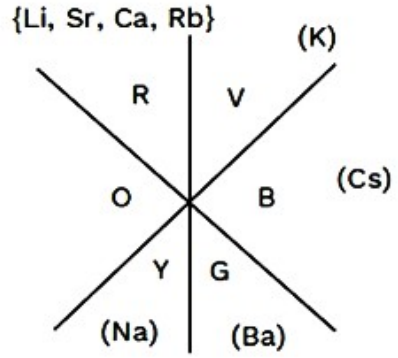
Due to low ionization energy they show characteristic colour to oxidizing flame. Due to energy provided by the flame electron oxidises to higher level and when it comes back to the ground state, radiation in visible range are observed.

With the help of atomic absorption spectroscopy or flame photometry alakil metals can be identified.

They energy provided may make them lose an electron from outer orbit as they have very low ionization energy.



Tips and Tricks



Lithium (Li)
Sharukh (Sr) Ke sath
Catrina (Ca)
Ranbeer (Rb) Re (K) sath
Barat (Ba) Main
Nache (Na)

Metal	Shade of color
Lithium	Carmine Red
Sr	Crimson Red
Ca	Brick Red
Rb	Red violet
K	Lilac (violet)
Ba	Apple Green
Na	Golden Yellow

Properties of the Alkali Metals

Electronic configuration	Li	Na	K	Rb	Cs	Fr
	[He] $2s^1$	[Ne] $3s^1$	[Ar] $4s^1$	[Kr] $5s^1$	[Xe] $6s^1$	[Rn]$7s^1$
Atomic Number	Li	Na	K	Rb	Cs	Fr
	3	11	19	37	55	87
Atomic mass (g mol^{-1})	Li	Na	K	Rb	Cs	Fr
	6.94	22.99	39.10	85.47	132.91	(223)
Ionization enthalpy/kJ mol^{-1}	Li	Na	K	Rb	Cs	Fr
	520	496	419	403	376	375
Metallic radius/ pm	Li	Na	K	Rb	Cs	Fr
	152	186	227	248	265	-
Ionic radius/pm	Li	Na	K	Rb	Cs	Fr
	76	102	138	152	167	(180)
Hydration enthalpy/kJ mol^{-1}	Li	Na	K	Rb	Cs	Fr
	−506	−406	−330	−310	−276	-
Density/gcm^{-3}	Li	Na	K	Rb	Cs	Fr
	0.53	0.97	0.86	1.53	1.90	-
m.p./K	Li	Na	K	Rb	Cs	Fr
	152	186	227	248	265	-
b.p/K	Li	Na	K	Rb	Cs	Fr
	1615	1156	1032	961	944	-
Standard Potentials for (M^+/M)	Li	Na	K	Rb	Cs	Fr
	−3.04	−2.714	−2.925	−2.930	−2.927	-

Chemical Properties

A low ionization energy and large size make alkali metals highly reactive. Their reactivity increases down the group.

(1) Reactivity towards air (N_2 + O_2)

With dry air they form oxides which further react with moisture to form hydroxide.

Reaction with N_2

$$N_2 \longrightarrow N \longrightarrow N^{-3}$$

$$\frac{1}{2}\left\{\begin{matrix}\text{Bond}\\ \text{dissociation}\\ \text{energy}\end{matrix}\right\}$$

$EA_1 + EA_2 + EA_3$ {Electron Affinity}

EA_1 of N → Endothermic {p^3 half filed}

EA_2 and EA_3 endothermic

To Convert $N \xrightarrow[\text{energy}]{\text{high}} N^{-3}$

Metals with high lattice energy can form nitride's.

$Li + N_2 \rightarrow Li_3N$

$\underbrace{\text{Na, K, Rb, Cs}}_{\Downarrow}$

Donot form nitride

{Cannot compensate the required energy in the formation of nitride ion due to large size.}

Reaction with oxygen

$$O_2 \xrightarrow{+e^-} O_2 \xrightarrow{+e^-} O_2^{-2} \xrightarrow{+e^-} O^{-2}$$

Superoxide Peroxide Normal oxide

exo endo More endothermic

From $O_2 \rightarrow O^{-2}$ required energy increases in every step

Li (High lattice energy) + $O_2 \rightarrow Li_2O$

(Small size) (major product)

Points to remember

To form ionic compounds, lattice energy plays an important role.

$Li + air \rightarrow \underset{\text{(major)}}{Li_3N} + \underset{\text{(minor)}}{Li_2O} + Li_2O_2$

$Na + air \rightarrow \underset{\text{(major)}}{Na_2O_2} + \underset{\text{(minor)}}{NaO_2}$

$$\left.\begin{matrix}K\\ Rb\\ Cs\end{matrix}\right\} + air \rightarrow \underset{\text{(major)}}{MO_2} + \underset{\text{minor}}{M_2O_2}$$

{M = k, Rb, Cs}

$Na + O_2 \rightarrow$ Na_2O_2 (major product)

$\left.\begin{matrix} K \\ Rb \\ Cs \end{matrix}\right\} + O_2 \rightarrow MO_2$ {superoxide} (major product)

{M = k, Rb, Cs]]

Q1 **Find the oxidation state (O.S.) of K in KO_2?**

Sol: The superoxide species is represented as O_2^-; since the compound is neutral, therefore, the oxidation state of potassium(K) is +1.

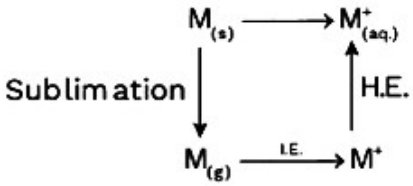
(2) Reactivity towards water:

A highly reactive metal like alkali metals librate hydrogen gas and form hydroxide of metal with water.

$$2M + 2H_2O \rightarrow 2M^+ + 2OH^- + H_2 \quad (M = \text{alkali metal})$$

Even though it has most negative E° value, Lithium reacts with water (H_2O) is not as vigorous as that of sodium (Na) which has the least negative E° value among first group. This is because of its tiny size and very great hydration energy. Remaining elements of the group react explosively with water (H_2O). They react with proton donors like gaseous ammonia, alcohol and alkynes.

$$\text{Metal} + H_2O \longrightarrow M^+OH^- + H_2\uparrow$$



As ionization energy decreases, rate of reaction increases

$\therefore$ Kinetic reactivity : Li < Na < K < Rb < Cs

Caesium reacts vigorously or even explosively with water as compared to lithium. sodium also reacts violently with water.

Thermodynamic reactivity depend upon net energy. net energy depend upon sublimation, ionization energy and hydration energy. Therefore, thermodynamic reactivity :

Na < K < Rb < Cs < Li

Hydration enthalpy $\times \frac{1}{\text{size}}$

Lithium being small in size has high ionization energy. Therefore, lithium has more thermolytic reactivity as compared to others.

(3) Reactivity towards halogens and dihydrogen:

With H_2 they(alkali metals) react at about 673K (lithium(Li) at 1073K) and form metal hydrides. First group hydrides are ionic solids and they have high M.P.(melting points). The stability of hydrides decreases down the group.

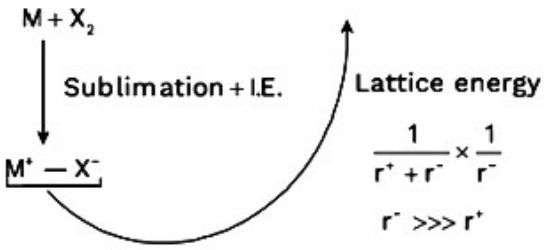
$$2M + H_2 \rightarrow 2M^+H^-$$

The reaction of first group metals with halogen results ionic halides, M^+X^-. This also a vigorous reaction. Due to small size, high polarising power of Li^+, the Lithium halides are covalent. Due to high polarizability of I^-, LiI is highly covalent among all Lithum halides.

$$2M + X_2 \rightarrow 2MX(s) \text{ ,Where, } X = F, Cl, Br, I, H$$

$-\Delta H_f$ = release energy

$M + X_2$



$$\text{Ionization energy} \times \frac{1}{\text{release energy}}$$

Order of release energy

MF : LiF > NaF > KF > RbF > CsF] depend on lattice energy

MCl, MBr, MI, MH } LiY < NaY < KY < RbY < CsY] dependon I.E.

(4) Reducing nature:

Based on the values of their reduction potentials we can conclude that the alkali metals are strong reducing agents.

Standard Potentials for (M^+/M)	Li	Na	K	Rb	Cs	Fr
	−3.04	−2.714	−2.925	−2.930	−2.927	-

The standard electrode potential ($E^{\ominus}$) which calculate the reducing power represents the overall change

$M(g) \rightarrow M^{+}(g) + e^{-}$ ionization enthalpy

$M(s) \rightarrow M(g)$ sublimation enthalpy

$M^{+}(g) + H_2O \rightarrow M^{+}(aq)$ hydration enthalpy

Lithium(Li) has the highest hydration enthalpy, due to very small size which is the reason for its high negative $E^{\ominus}$ value.

Hence, Li is highly reducing and Na is least reducing in nature.

Q2 **Based of reduction potential value arrange following according to reducing power: I^{-}, Ag, Cl^{-}, Li, Na**
if $E^{\ominus}$ for Cl_2/Cl^{-} is +1.36, for I_2/I^{-} is + 0.53, for Ag^{+}/Ag is +0.79, Na^{+}/Na is –2.71 and for Li^{+}/Li is –3.04.

Sol: The correct order is $Li > Na > I^{-} > Ag > Cl^{-}$

(5) Solutions in liquid ammonia :

$$M + \text{liq. } NH_3 \longrightarrow M^{+}_{(\text{ammoniated})} + e^{-}_{(\text{ammoniated})}$$

In the above reaction, solvent is pure NH_3 and solute is metal.

Bond polarity :

$H_2O > NH_3$

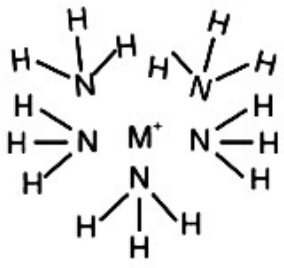
$H_2O \overset{K_1}{\rightleftharpoons} H^{+} + OH^{-}$

$NH_3 \overset{K_2}{\rightleftharpoons} NH_3 + H^{+}$

$K_1 >> K_2$

The extent of self-ionization in water is much more than in ammonia.

$M \longrightarrow M^{+} + e^{-}$

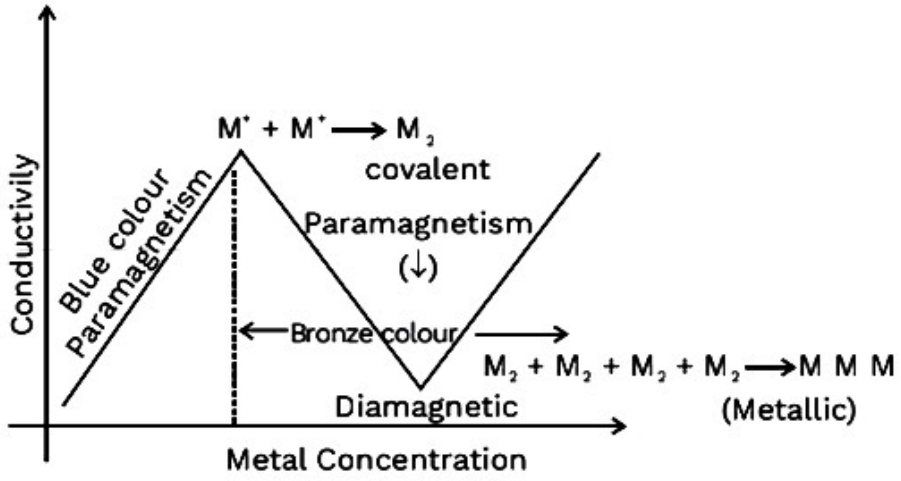


$M^{+}_{(\text{ammoniated})}$ ions shows conductivity.

Deep blue solution is formed on putting alkali metals in liquid ammonia. These solutions are which are conducting in nature due to availability of free solvated electrons. These electrons are called ammoniated electrons. These electrons absorve light in visible range owing to the blue colour of solution. This solution is paramagnetic and librate hydrogen slowly creating amide as a result.

$$\underset{V_1}{M} + \underset{V_2}{(x+y)NH_3} \longrightarrow \underbrace{[M(NH_3)_x]^+ + [e(NH_3)_y]^-}_{V_3}$$

$V_3 > V_1 + V_2$



$NH_3 \xrightarrow{\Delta} NH_2^- \; H^+$

$M + \text{liquid } NH_3 \xrightarrow{\Delta} H_2 \uparrow + M^+NH_2^-$

Colourless solution is obtained in above reaction. The solution obtained is diamagnetic and conducting in nature.

$M(\text{alkali metals}) + C_2H_5OH \rightarrow H_2 \uparrow + C_2H_5O^-M^+$

$M + NH_3(g) \longrightarrow H_2\uparrow + NH_2^-M^+$

(6) Reaction with H_2:

Lithium reacts with N_2 to form ionic lithium nitride Li_3N. Other metals of this group do not react with nitrogen..

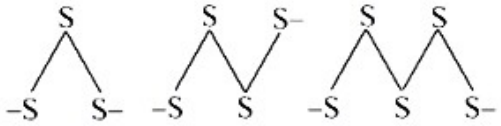
(7) Reaction with N_2:

Only Lithium reacts with N_2 to form ionic lithium nitride Li_3N.

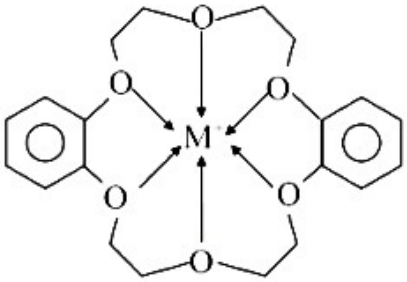
$$3Li + \frac{3}{2}N_2 \rightarrow Li_3N$$

(8) Sulphides:

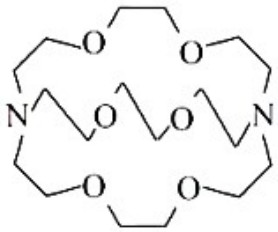
First group elements react with Sulphur(S) and results sulphides like Na_2S and polysulphines like Na_2S_n (n = 2, 3, 4,5 or 6). Zig-zag chains of sulphur atoms form the polysulphide ions.



(9) Crown Ethers and Cryptands:



Dibenzo- 18-Grown-6
[Na (Cryptand- 222)]$^+$ Na$^-$
[s$^+$ (Cryptand- 222)]



Cryptand - 222
[Contains Na$^-$(sodide ion)]
[(Cryptand-222)e$^-$] [electride]

Q3 **List three general characteristics of al kali elements which distinguish them from the elements of the other blocks.**

Sol: (a) The ns^1 configuration
(b) High E°_{OP} values and thus strong reducing agent
(c) Soluble in liquid NH_3 and give highly conducting blue coloured solution.

Q4 **Name the alkali metals which form superoxide's when heated in excess of air.**

Sol: K, Rb and Cs form superoxide's as KO_2, RbO_2 and CsO_2 respectively.

Q5 **The low density alkali metal which floats on water is?**

Sol: Lithium

Q6 **List three properties of lithium in which it differs from the rest of the alkali metals.**

Sol: (a) It forms deliquescent salts
(b) It is much harder than other alkali metals
(c) It is less reactive but strongest reducing agent among all alkali metals.

Q7 **Name the chief factor responsible for the anomalous behaviour of lithium.**

Sol: The charge/radius ratio is greatest for lithium.

Q8 **Mention the general trends in Group I with increasing atomic number with respect to:**

Sol: (A) Density
(B) Melting point
(C) Decreases
(D) Decreases

Q9 **Why are alkali metals kept in paraffin or kerosene?**

Sol: Alkali metals are highly sensitive towards air and water and so are kept in kerosene or paraffin oil.

Q10 **Themselves being paramagnetic why, generally, alkali metals form diamagnetic salts ?**

Sol: Alkali metals have one unpaired electron (ns^1) and are paramagnetic. However, during the salt formation, the unpaired electron is lost metals to other atom forming anion. The salt has all electrons are shows diamagnetic nature.

Q11 **Aqueous solution of iodine becomes colourless on adding excess of sodium hydroxide solution. Why ?**

Sol: Iodine present in solution reacts with NaOH and thus loses its colour.

$$I_2 + \underset{(cold)}{2NaOH} \longrightarrow NaI + NaIO + H_2O$$

$$3I_2 + \underset{(hot)}{6NaOH} \longrightarrow 5NaI + NaIO_3 + 3H_2O$$

Q12 **Why are alkali metals used in photoelectric cells ?**

Sol: Due to lower ionisation energy, removal of photo-electrons takes place easily even in visible radiation.

Q13 **Why are K and Cs rather than Na and Li used in photoelectric cells ?**

Sol: Due to lower ionisation energy of K and Cs, the photoelectric emission of electron takes place easily.

Q14 **Cesium is most reactive among the alkali metals, explain.**

Sol: Due to lower ionisation energy. It possesses more electropositive nature.

Q15 **Alkali metals are soft metals, explain.**

Sol: Alkali metals have largest atomic radii in their periods. The degree of metallic bonding in alkali metals is lower as metallic bonding with increase in atomic size.

Q16 **Why Li forms covalent halides?**

Sol: Li^+ ion due to smaller size other alkali metal cations exerts stronger polarizing power towards anion and therefore develops covalent in the molecule (Fajan rule)

Q17 **Super oxides of alkali metals are coloured and paramagnetic in nature. Why?**

Sol: Super oxides contain a three electron bond which makes its paramagnetic and coloured.

Q18 **Chemical reactivity of alkali metal increases from Li to Cs.**

Sol: Ionisation energies decrease from Li to Cs showing highest reactivity.

Q19 **Why is colour of the flame imparted by sodium salt different from that potassium salts ?**

Sol: The excitation in Na occurs from 3rd shell, whereas in K from 4th shell.

Q20 **Which out of Na and K has high m.pt. and density ?**

Sol: **(A) m.pt.-** Na; As the atomic size increases, interparticle distance between atoms in crystal lattice increases and the interparticle forces operating in between them decreases. Thus m.pt. decreases down the group.
(B) Density – Na

Q21 **Alkali metal salts imparts colour to flame.**

Sol: Due to low ionization energy they show characteristic colour to oxidizing flame. Due to energy provided by the flame electron oxidises to higher level and when it comes back to the ground state, radiation in visible range are observed.
These different colours are due to different ionisation energy of alkali metals.
The energy released is the case of Li and increases in the order :

Energy released : Li < Na < K < Rb < Cs

λ released : Li > Na > K > Rb > Cs

Frequency released : Li < Na < K < Rb < Cs

Metal	Shade of color
Lithium	Carmine Red
Sr	Crimson Red
Ca	Brick Red
Rb	Red violet
K	Lilac (violet)
Ba	Apple Green
Na	Golden Yellow

Q22 **LiF, LiCl, LiBr and LiI in increasing covalent character and explain the reason:**

Sol: LiF < LiCl < LiBr < LiI, Larger is anion (I^-) more is deformation by cation and more is covalent character. (Fajan's rule).

Q23 **Lithium halides have a greater covalent character than other metal halides of the group. Why?.**

Sol: The small size of Li^+ gives it a huge polarising power. The electron cloud of larger halide ion gets distorted and this leads to covalency.

Q24 **Salts of lithium having small anions are more stable than the salts having larger anions. Explain.**

Sol: The salts of Li^+ with small anions have high lattice energy and thus very stable while with large anion, the lattice energy decreases and the salts become less stable.

Q25 **What are wrap around complex compound of alkali metals ?**

Sol: Alkali metals have minimum tendency to form complex compounds, however polydentate crown ligand forms stable complexes with alkali metals example, M^+ with 18-crown-6-ether forms wrap around complex.

Important Compounds of Alkali Metals

Hydroxides and Oxides

On reaction with excess air under controlled conditions M_2O_2, MO_2 and M_2O type of compounds can be formed. Li mainly forms oxides, Na peroxide and K, Rb & Cs can form superoxides as well.

Similar sized ions form a better packing owing a very good stability of ionic salts. Since peroxide

Points to remember

Peroxides: Colourless in pure form.
Superoxides: orange or yellow in colour, paramagnetic.
Sodium peroxide(Na_2O_2) a common oxidizing agent in inorganic chemistry.

and superoxide ions are large in size they will be better packed with metal ions of higher sizes. This explains better stability of peroxides and super oxides for metal lying below in the group. The hydrolysis of these oxides form the hydroxides. The reactions are as follow:

$M_2O + H_2O \rightarrow 2M^+ + 2OH^-$

$M_2O_2 + 2H_2O \rightarrow 2M^+ + 2OH^- + H_2O_2$

$2MO_2 + 2H_2O \rightarrow 2M^+ + 2OH^- + H_2O_2 + O_2$

Q26 **Explain reason paramagnetic behavior of KO_2 ?**

Sol: The ion O_2^- is paramagnetic with one upaired electron in π^*2p molecular orbital while K^+ ion is diamagnetic. Hence due to O_2^-, KO_2 is paramagnetic in nature.

Key Points to Remember

Stability of oxides decreases top to bottom

$Li_2O > Na_2O > K_2O > Rb_2O > Cs_2O$

Oxide

O^{2-} (Monoatomic anion)

Thermal stability α Lattice energy

Superoxide $\left(O_2^-\right)$, Peroxide $\left(O_2^{2-}\right)$

(Polyatomic)

Thermal stability $\alpha \frac{1}{\text{polarising power}}$

Stability:

$Li < Na < K < Rb < Cs$

Uses:

Na_2O_2 and KO_2 are source of O_2 and act as CO_2 absorber. They are used in submarines and spacecrafts.

$KO_2 \xrightarrow{\Delta} K_2O + O_2$

$Na_2O_2 \xrightarrow{\Delta} Na_2O + O_2$

$O_2 \xrightarrow{e^-} O_2^- \xrightarrow{e^-} O_2^{2-} \xrightarrow{e^-} O^{-2}$

Key Reactions for Hydroxide

Thermal Stability: LiOH < NaOH < KOH < RbOH < CsOH

Solubility: LiOH < NaOH < KOH < RbOH < CsOH

Basic Nature: LiOH < NaOH < KOH < RbOH < CsOH

As we move from top to bottom, tendency to given ^-OH ions increases.

Sodium Oxide (Na_2O):

Preparation :

(i) In controlled access to oxygen or air at 180°C Sodium oxide is formed. The extra Sodium is distilled out in vacuum.

$$2Na + \frac{1}{2}O_2 \xrightarrow{180^\circ} Na_2O$$

(ii) Thermal exposure of sodium peroxide(Na_2O_2), nitrate or nitrite to sodium metal leads to formation of sodium oxide.

$$Na_2O_2 + 2Na \longrightarrow 2Na_2O$$
$$2NaNO_2 + 6Na \rightarrow 4Na_2O + N_2$$
$$2NaNO_3 + 10Na \rightarrow 6Na_2O + N_2$$

Properties :

(i) It is white amorphous mass.

(ii) At 400°C it gets converted into sodium peroxide and sodium

$$2Na_2O \xrightarrow{400^\circ} Na_2O_2 + 2Na$$

(iii) It forms caustic soda as result of violent reaction with water.

$$Na_2O + H_2O \rightarrow 2NaOH$$

Sodium Peroxides (Na_2O_2):

Preparation:

When sodium is heated with dry air or oxygen at 300°. The reaction must be carried out in moisture free scenario to avoid formation of hydroxide. Also presence of CO_2 is avoided to avaoid acid base reaction with basic peroxide.

$$2Na + O_2 \rightarrow Na_2O_2$$

Properties:

(i) It is a pale yellow solid. It becomes white in air forming a film of Na_2CO_3 and NaOH.

(ii) In cold water (0°C) produces H_2O_2 but at room temperature produces O_2. In ice-cold mineral acids also produces H_2O_2.

$$Na_2O_2 + 2H_2O \xrightarrow{0^\circ C} 2NaOH + H_2O_2$$
$$2Na_2O_2 + 2H_2O \xrightarrow{25^\circ C} 4NaOH + O_2$$
$$Na_2O_2 + H_2SO_4 \xrightarrow{0^\circ C} Na_2SO_4 + H_2O_2$$

(iii) Its acid-base reaction with CO_2 is reason why it is used to purify confinrd spaces like submarines. The reraction results sodium carbonate and oxygen,

$2Na_2O_2 + 2CO_2 \rightarrow 2Na_2CO_3 + O_2$

(iv) It is an oxidising agent(O.A.) and oxidises NH_3, charcoal, CO, , SO_2.

$CO + Na_2O_2 \rightarrow Na_2CO_3$

$3Na_2O_2 + 2C \rightarrow 2Na_2CO_3 + 2Na$ [deposition of metallic sodium]

$SO_2 + Na_2O_2 \rightarrow Na_2SO_4$

$2NH_3 + 3Na_2O_2 \rightarrow 6NaOH + N_2$

(v) It contains peroxide ion $[-O-O-]^{-2}$

Uses:

(i) To manufacture H_2O_2, O_2

(ii) Purifying and oxygenating submarines

(iii) Laboratories as O.A.

Oxides of Potassium:

K_2O (White), K_2O_2(White), K_2O_3(Red), KO_2 (Bright yellow) and KO_3(Orange Solid)

Preparation:

(i) $2KNO_3 + 10K \xrightarrow{\text{heating}} 6K_2O + N_2$

$K_2O \xrightarrow{\text{heating}} K_2O$

(White) (Yellow)

$K_2O + H_2O \rightarrow 2KOH$

(ii) $2K + O_2 \xrightarrow[\text{air at } 300^\circ C]{\text{Controlled}} K_2O_2$

[**Note:** Similar with sodium peroxide(Na_2O_2)]

(iii) On treating K in liquid NH_3 with O_2 yields oxides K_2O_2 (white), K_2O_3 (red) and KO_2 (deep yellow) i.e

K in liq. $NH_3 \xrightarrow{O_2}$ K_2O_2 (white) $\rightarrow$ K_2O_3 (red) $\rightarrow$ KO_2 (yellow)

white red yellow

KO_2 reacts with water(H_2O) and produces hydrogen peroxide(H_2O_2)and O_2 both

$2KO_2 + 2H_2O \xrightarrow{0^\circ C} 2KOH + H_2O_2 + O_2$

KO_3 : KOH + O_3 (ozonised oxygen) $\xrightarrow{-10^\circ \text{to} -15^\circ C}$ KO_3

(Dry powdered) (orange solid)

Halides

MX, (X=F, Cl, Br, I) are high melting, colourless crystalline solids.
Made by the reaction of the appropriate hydroxide ,oxide or carbonate with aqueous hydrohalic acid (HX).
Halides have high negative enthalpies of formation

Key Concepts

In F^- Solubility: LiF < NaF < KF < RbF < CsF

In I^- Solubility: LiI > NaI > KI > RbI > CsI

Lithium Pralides $\left(Cl^-, Br^-, I^-\right)$ are predominantly covalent. So, they are also soluble in acetone.

Salts of Oxo-Acids

Compound of an atom, connected with an oxo group (=O), capable of furnishing H^+ ion attached with hydroxy group is known as oxy acid. E.g. H_2SO_4 $(O_2S(OH)_2)$, H_2CO_3 $(OC(OH)_2$ etc.

Alkali metal reacts with all oxo acids to form salts.
Generally, all salts of alkali metals are soluble in water.
They form thermally stable carbonates (M_2CO_3), $(MHCO_3)$
As the size increases down the group, effective packing with carbonates and bicarbonate anions increases and hence the stability increases.

Having very small size of Li^+ as compared to CO_3^{2-}, Lithium carbonate(Li_2CO_3) is not effectively packed, leading to its low thermal stability. Its hydrogen carbonate does not occur as a solid.

(a) Thermal stability decreases from top to bottom

$Li_2CO_3 \longrightarrow Li_2O_2 + CO_2 + O_2$

$Na/K/Rb/Cs \xrightarrow{\Delta} X$ [Na/K/Rb/Cs has high thermal stability)

$LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2 + O_2$

(b) Solubility: Solubility of $Co_3^{2-}, HCO_3^-, SO_4^{2-}, SO_3^{2-}$ increases

ANOMALOUS PROPERTIES OF LITHIUM

Li among all alkali metals show anomalous behavior because of exceptionally tiny size of its atom and ion, high polarizing power (i.e., charge/ radius ratio) and covalent character of lithium compounds making them soluble in organic solvents.

Points of Difference among Lithium and other Alkali Metals

Lithium(Li) has high melting point(M.P.) and boiling point(B.P.) than the other elements of the group.

Li has lowest reduction potential among all, owing to its high reducing power, though it is least reactive element in the group. On reaction with air it forms, Lithium oxide Li_2O and the nitride, Li3N unlike other alkali metals.

Lithium chloride(LiCl) forms crystalline hydrate LiCl. $2H_2O$ unlike other elements of the group.

Bicarbonate of Li is not found in solid state while other elements of this group form solid bicarbonates.

Only Li in the group forms ethynide on reaction with ethyne.

While other metals form nitride, Li forms oxide on decomposition of its nitrate

$4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$

$2NaNO_3 \rightarrow 2NaNO_2 + O_2$

Points of Similarities among Lithium and Magnesium:

Both Li and Mg are harder and lighter than other elements in the respective groups.
Both form nitride with air and their oxides are less soluble in water than their respective group elements. They do not form super oxides.
Carbonates of both the elements are thermally less stable.
Both form alcohol soluble chlorides.

SOME IMPORTANT COMPOUNDS OF SODIUM (Na)

- Sodium carbonate (Na_2CO_3),
- Sodium hydroxide (NaOH),
- sodium chloride(NaCl)
- sodium bicarbonate ($NaHCO_3$)

Sodium Carbonate (Washing Soda), Na2CO3·10H2O (Solvay Process)

- **Method of preparation: Solvay Process**

  Solubility of sodium bicarbonate ($NaHCO_3$) is low and it gets precipitated in the reaction NaCl with NH_4HCO_3. NH_4HCO_3 made by passing CO_2 through concentrated solution of NaCl saturated with NH3, where $(NH_4)_2CO_3$ followed by formation of NH_4HCO_3.

  $2NH_3 + H_2O + CO_2 \rightarrow (NH_4)2\ CO_3$

  $(NH_4)_2\ CO_3 + H_2O + CO_2 \rightarrow 2NH_4HCO_3$

  $NH_4HCO_3 + NaCl \rightarrow NH_4Cl + NaHCO_3$

- The $NaHCO_3$ formed in above reaction is heated to get Na_2CO_3.

  $2NaHCO_3 \rightarrow Na_2CO + CO + H_2O$

- Ammonium (NH_3) is recovered by treating NH_4Cl with $Ca(OH)_2$.

  $2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + CaCl_2 + H_2O$

Solvay is not used to the manufacture of potassium carbonate (K_2CO_3) because of high solubility of $KHCO_3$ in ammonium hydrogen carbonate to a saturated solution of potassium chloride (KCl).

Properties of Na2CO3 :

Na_2CO_3, washing soda, is a white crystalline solid, soluble in water. On heating, the decahydrate (a compound with 10 molecules of water) loses its water of crystallisation to form monohydrate.

$$Na_2CO_2 \rightarrow 10H_2O \xrightarrow{375K} Na_2CO_3 \rightarrow H_2O + 9H_2O$$

The Na_2CO_3.H2O becomes completely anhydrous and changes to a white powder called soda ash at above 373K.

$$Na_2CO_3 \rightarrow H_2O \xrightarrow{>373K} Na_2CO_3 + H_2O$$

Uses:

It is used for laundering, softening and cleaning, manufacture of soap, borax, glass and caustic soda. It is utilized in paper, paints and textile industries. It is an important laboratory reagent for qualitative and quantitative analysis.

Sodium Chloride (NaCl)

Main source of sodium chloride (NaCl) is sea water which contains 2.7 to 2.9% by mass of the salt.

Uses :

It is used as a common salt or table salt. It is utilized for the preparation of Na_2O_2, NaOH and Na_2CO_3. It is used to prepare freezing mixture in laboratory and it is a common reagent for melting ice and snow on road.

Sodium Hydroxide (NaOH / Caustic Soda)

- Prepared by the electrolysis of NaCl, brine solution, using a mercury cathode and a carbon anode.
- Sodium amalgam is formed at the cathode when sodium metal discharges and Cl2 is evolved at the anode.

  Cathode : $Na^+ + e^- \xrightarrow{Hg} Na-$ amalgam

  Anode : $Cl^- \rightarrow \frac{1}{2} Cl_2 + e^-$
- Amalgam with H2O gives sodium hydroxide (NaOH) and hydrogen gas.

  $2Na\text{-amalgam} + 2H_2O \rightarrow 2NaOH + 2Hg + H_2$
- NaOH is a white, glowing solid.
- Melting point of NaOH is 591 K. It is highly soluble in water and gives a strong alkaline solution. Crystals of NaOH are deliquescent.
- The sodium hydroxide(NaOH) solution at the surface reacts with the CO_2 in the atmosphere to form Na_2CO_3.

  Uses: It is utilized in manufacture of soap, paper, artificial silk and a number of chemicals, petroleum refining, purification of bauxite, textile industries for mercerizing cotton fabrics.

Sodium Hydrogen carbonate ($NaHCO_3$ / Baking Soda)

- Saturated Na_2CO_3 solution with excess CO_2 gives $NaHCO_3$ which separates out of solution being less soluble.
- $Na_2CO3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$
- It is called baking soda. It leaves bubbles of CO_2 Making cakes or pastries fluffy.
- $NaHCO_3$ is a mild antiseptic for skin infections. It is utilized in fire extinguishers.

Q27 **NaCl in water forms a neutral solution whereas Na_2CO_3 in water gives alkaline solution. Why ?**

Sol: NaCl is a salt of strong acid and strong bases and thus, does not hydrolyse, whereas Na_2CO_3, a salt of strong base and weak acid (H_2CO_3), gets hydrolysed.

$$CO_3^{2-} + 2H_2O \rightleftharpoons H_2CO_3 + 2OH^-$$

Q28 **When sodium carbonate is added to a solution of aluminium salt, aluminium hydroxide is precipitated. Why ?**

Sol: $Al_2(CO_3)_3$ and $Al(OH)_3$ are formed which are insoluble in water.

Q29 **List two uses of.**
(A) Caustic soda **(B) Sodium carbonate** **(C) Quick lime**

Sol: (A) In soap industry, in petroleum industry.
(B) In glass and soap industry, in paper making
(C) As drying agent, as a constituent of mortar.

Q30 **Why are alkali metals not found in uncombine state nature ?**

Sol: Alkali metals are highly reactive because of low IE values and are thus found in nature only in the combined state.

Q31 **Explain rest reactivity of Na as compared to K ?**

Sol: Due to low IE of K it is more reactive than Na.

Q32 **To form alkali metals molten solution instead of aq. solution of metal salts is used. why?**

Sol: On electrolysis of aqueous sodium of alkali metal salt (containing metal cation, H^+, OH^- and other anion), the alkali metal cations due to higher discharge potential than H^+ do not discharge at cathode. On the contrary, it is the H^+ ion which is discharge at cathode to give H_2.
However, the electrolysis of fused metal salt, liberates metal at cathode due to the discharge of metal cation at cathode.

Q33 **Why does sodium form Na^+ ion and not Na^{2+} ion ?**

Sol: Na^+ ion acquires stable noble gas configuration after the removed of electron from 3s-orbital and removed of electron from 2p-orbital of stable Na^+ require a lot of energy.

$$Na \xrightarrow{+520\,kJ\,mol^{-1}} Na^+ \xrightarrow{-7296\,kJmol^{-1}} Na^{2+}$$

Q34 **Unlike Na_2CO_3, K_2CO_3 cannot be prepared by Solvay ammonia soda process. Why ?**

Sol: Unlike $NaHCO_3$, $KHCO_3$ is highly soluble in water and cannot be separated out to heat up and get $K2CO_3$. This is the main reason why Solvay process cannot be used for $K2CO_3$

Q35 **Though placed in different groups of periodic table, lithium and magnesium resemble each other. Why ?**

Sol: Both Li and Mg have nearly same charge-size ratio and other closer periodic properties and thus resemble in many other properties (diagonal relationship).

Q36 **What is the composition and action of baking powered ?**

Sol: It is mixture of potassium hydrogen tartrate and sodium bicarbonate. On heating (baking). $CO_{2(g)}$ is evolved which causes the bread to swell.

Q37 **$LiClO_4$ is much more soluble in H_2O than $NaClO_4$, why ?**

Sol: Due to strong solvation of smaller Li^+ ion, much energy is released which makes $LiClO_4$ more soluble.

Q38 **LiH is stable to approximately 900°C while NaH decomposes at 350°C why ?**

Sol: Due to small size of Li^+, LiH has higher lattice energy than NaH.

Q39 **In preparation of LiH by action of H_2 on heated Li, heating is discontinued after reaction has begin, why ?**

Sol: Since the heat of reaction is sufficient to maintain the temperature high enough for reaction to occur.

GROUP 2 ELEMENTS : ALKALINE EARTH METALS

1. Beryllium(Be),
2. Magnesium(Mg),
3. Calcium(Ca),
4. Strontium(Sr),
5. Barium(Ba)
6. Radium(Ra).

Group 2 elements follow alkali metals in the periodic table. These (except beryllium(Be)) are called alkaline earth metals. The first element beryllium(Be) differs from the rest of the members and exibit diagonal relationship to aluminium(Al). The physical and atomic properties of the alkaline earth metals are given below.

Electronic Configuration (E.C.)

These elements have 2 electrons in the s-orbital of the valency shell. Their common electronic configuration may be denoted as [noble gas] ns^2. Just like alkali metals, the compounds of these elements are also predominantly ionic.

Element	Symbol	Electronic configuration
Beryllium	Be	$1s^2 2s^2$
Magnesium	Mg	$1s^2 2s^2 2p^6 3s^2$
Calcium	Ca	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Strontium	Sr	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^2$

Element	Symbol	Electronic configuration
Barium	Ba	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 6s^2$
Radium	Ra	$[Rn]\ 7s^2$

Physical and Atomic Properties of the Group 2 Elements (Alkaline Earth Metals)

Property	Beryllium Be	Magnesium mg	Calcium Ca	Strontium Sr	Brium Ba	Radium Ra
Atomic Number	4	12	20	38	56	88
Atomic mass ($g\ mol^{-1}$)	9.01	24.31	40.08	87.62	137.33	222.03
Electronic configuration	$[He]\ 2s^2$	$[Ne]\ 3s^2$	$[Ar]\ 4s^2$	$[Kr]\ 5s^2$	$[Xe]\ 6s^2$	$[Rn]7s^2$
Ionization enthalpy (I)/$kJ\ mol^{-1}$	899	737	590	549	503	509
Hydration enthalpy (II)/$kJ\ mol^{-1}$	−2494	−1921	−1577	−1443	−1305	-
Metallic radius/ pm	111	160	197	215	222	-
Ionic radius M^+/ pm	31	72	100	118	135	148
m.p./K	1560	924	1124	1062	1002	973
b.p/K	2745	1363	1767	1655	2078	(1973)
Density/gcm^{-3}	1.84	1.74	1.55	2.63	3.59	(5.5)
Standard Potentials E^-/V for (M^+/M)	−1.97	−2.36	−2.84	−2.89	−2.92	−2.92

Ionization Enthalpy

Be > Mg > Ca > Sr > Ba

Down the group Ionization Enthalpy (I.E.) decreases due for to increase in size

Hydration Enthalpies

Like group 1 (alkali metal) ions, the hydration enthalpies of group 2 (alkaline earth metal) ions reduce with increase in ionic size down the group.

$Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

Points to remember

They have high hydration energy as compared to respective first group elements due to high effective nuclear charge and small size. Their compounds are hydrated extensively as compared to that of first group. This is the reason why MgCl2 and $CaCl_2$ occur as $MgCl_2.6H_2O$ and $CaCl_2 \cdot 6H_2O$ while KCl and NaCl do not form such hydrates.

Physical Properties

They are soft, (relatively harder than first group) silvery white and lustrous. Be and Mg are grayish in colour. Their small size, as compared to first group, imparts relatively high melting and boiling points to them. Their low ionization energy makes them high electropositive, yet they are less electropositive as compared to first group. The electropositive character improves down the group from Beryllium(Be) to Barium (Ba).

Flame Colour
Calcium(Ca) - brick red
strontium(Sr) - crimson and
barium(Ba)- apple green
Beryllium(Be) and magnesium(Mg) do not impart any colour to the flame.

Chemical Properties

Reactivity of these elements increases on going down the group.

Reactivity → Alkaline earth metals < Alkali metals

(i) Reactivity towards air :

Beryllium (Powdered) and Magnesium(Mg) brilliantly react to form Oxide(BeO/MgO) and nitride($Be3N_2$/ Mg_3N_2) with air while Ca, Sr, Ba are readily attacked by air to form the oxide and nitride.

(ii) Reactivity towards water.

Similar to that of first group second group elements also form hydroxide with water releasing H_2 gas. They are relatively less reactive towards water as compared to first group. Be is inert towards water. The products are basic in nature.

$$M + 2H_2O \xrightarrow{\Delta} M(OH)_2 + H_2$$

Formation of a protective layer makes Mg less readily reactive towards water unless layer is removed amalgamating with Hg. Other metals react quite readily (Ca, Sr, Ba).

(iii) Reactivity towards the halogens:

All the alkaline earth metals come together with halogen (X = F, Cl, Br, I) at elevated temperatures forming their halides.

$M + X_2 \rightarrow MX_2$

Thermal decomposition of $(NH_4)_2BeF_4$ is the most excellent route for the formulation of $BeCl_2$, and BeF_2 is conveniently made from the oxide.

$$BeO + C + Cl_2 \xrightarrow{600-800} BeCl_2 + CO$$

(iv) Reactivity towards hydrogen:

While other elements of second group form hydride (MH_2) on direct heating with hydrogen, BeH_2 is prepared by reaction $BeCl_2$ with $LiAlH_4$.

$2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$

(v) Reactivity towards acids:

Just like other reactive metals alkali metals liberate H2 on reaction with acids.

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

Be being amphoteric reacts with bases like NaOH as well to form hydrogen.

$$Be + 2NaOH \rightarrow Be(OH)_2 \xrightarrow{excess\text{-}NaOH} [Be(OH)_4]^{2-}$$

(vi) Reducing nature:

Large negative reduction potential makes alkaline earth metals strong reducing agents. Alkali metals still have high reducing tendency than alkaline earth metals.

(vii) Solutions in liquid ammonia:

Alkaline earth metals dissolve in liquid ammonia(liq. NH_3) to give deep blue black solutions producing ammoniated ions.

$(x + y)\ NH_3 + M \rightarrow [M(NH_3)_x]^{2+} + 2[e(NH_3)_y]^-$

From the solutions, the ammoniates, $[M(NH_3)_6]^{2+}$ can be regained.

Uses

Beryllium (Be) :

Beryllium is utilized in the manufacture of alloys. High strength springs are made using Copper-beryllium alloys. Windows of X-ray tubes use Metallic beryllium.

Magnesium (Mg):

Magnesium is used in range of alloys with zinc, aluminium, manganese and tin. Light Magnesium-aluminium alloys are utsed in air-crafts. . Magnesium (powder and ribbon) is utilized in flash powders and bulbs, incendiary bombs and signals. A suspension of MgCO3 in water (known as milk of magnesia) is utilized as antacid in medicine. MgCO3 is an ingredient of toothpaste as well.

Calcium (Ca) :

Oxides which are difficult to reduce by C are reduced using calcium during extraction of metals. To remove air from vacuum Calcium (Ca) and barium (Ba) metals are used at elevated temperature.

Radium (Ra) :

Radium salts are utilized in radiotherapy, for example, in the treatment of cancer.

CHARACTERISTICS OF COMPOUNDS OF THE ALKALINE EARTH METALS

They are divalent and they mainly form ionic compounds. The compounds of second group are less ionic as compared to first group as they have high polarizing power due to their small size compared to respective first group element.

(i) Hydroxides and Oxides:

They form basic, ionic oxides having rock salt type structure with oxygen. Be is an exception in this manner. BeO is covalent, amphoteric substance. Covalent character is BeO is a result of high polarization and high ionization energy of Be.

The oxides with water form hydroxides

$$MO + H_2O \rightarrow M(OH)_2$$

$Mg(OH)_2$
$Ca(OH)_2$ | Solubility, thermal stability and the basic nature increases down
$Sr(OH)_2$
$Ba(OH)_2$ ↓

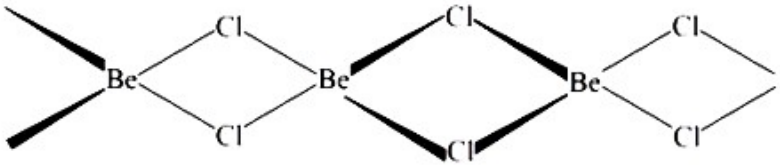
$Be(OH)_2$ being amphoteric reacts with both acids as well as base.

$$2HCl + Be(OH)_2 + 2H_2O \rightarrow [Be(OH)_4]Cl_2$$

$$2OH^- + Be(OH)_2 \rightarrow [Be(OH)_4]^{2-}$$

(ii) Halides:

All MX_2 are ionic in nature except that of BeX_2 (Covalent and soluble in organic solvents). $BeCl_2$ has a chain structure in the solid.



$BeCl_2$ vapor forms a dimer which dissociates into the linear monomer at high temperatures of the order of 1200 K.

$MgCl_2$
$CaCl_2$
$SrCl_2$
$BaCl_2$ ↓ Tendency to get hydrated decreases down the group.

Dehydration of hydrated bromides, chlorides, and iodides of calcium(Ca), strontium(Sr), barium(Ba) can be attained on heating; though, the equivalent hydrated halides of beryllium(Be) and magnesium(Mg) on heating suffer hydrolysis. The fluorides are comparatively less soluble than the chlorides due to their high-level lattice energy.

Salts of Oxoacids:

Carbonates: MCO_3 are generally insoluble in water

$MgCO_3$
$CaCO_3$
$SrCO_3$
$BaCO_3$ ↓ Solubility of carbonates decreases down the group.
Thermal stability increases down the group

All the carbonates produce carbon dioxide(CO2) on heating. BeCO3 is unstable and can sustain only in CO2 atmosphere.

Sulphates:

White solids and thermally stable. Due to large hydration energy of Be^{2+} and Mg^{2+} $BeSO_4$ and $MgSO_4$ are readily soluble in water; Solubility decreases down the group. Ca to Ba sulphates are less soluble in water.

Nitrates:

Their carbonates with nitric acid form nitrate. Due to decreasing hydration tendency down the group $Mg(NO_3)_2$ crystallises with 6 molecules of water, whereas $Ba(NO_3)_2$ crystallises as the anhydrous salt. Just like lithium nitrate all of them decompose on heating to give their oxide.

$2M(NO_3)_2 \rightarrow 2MO + 4NO_2 + O_2$ (Here 'M'→ alkaline earth metals)

ABNORMAL BEHAVIOUR OF BERYLLIUM (Be)

Due extremely small size and high ionization energy Be, the first member of the Group 2 metals, shows abnormal behaviour as compared rest of the members. It mainly forms covalent compounds due to high polarizing power as a result of tinny size. Its compounds get easily hydrolysed.

While other member exibit a coordination number of 6 Be only exibits a maximum of 4 coordination number as it has onll 4 orbitals available for bonding.

Unlike other elements Be form amphoteric oxides and hydroxides.

Diagonal Relationship among Beryllium (Be) and Aluminium (Al)

The ionic radius of Be2+ is similar to that of Al3+. Therefore Beryllium(Be) resembles aluminium in some ways. Just Like Aluminium, beryllium forms protective oxide layer on the surface which prevents it from attack of acids. Be(OH)2 dissolves in excess of base to produce a beryllate ion, [Be(OH)4]2– just as aluminium hydroxide produce aluminate ion, [Al(OH)4]–. Both form bridged bond covalent chlorides, soluble in organic solvents. They are common catalysts in Friedel Craft reaction.

Q40 **Like lithium in group I, beryllium shows anomalous behaviour in group II. Write three such properties of beryllium which make it anomalous in the group.**

Sol: (a) Be does not show co-ordination number more than four.
(b) The oxide and hydroxides of Be are amphoteric in nature.
(c) $BeCl_2$ has bridged chloride structure in vapour phase.

Q41 **IE_3 of Mg is very high as compared to its IE_2 value.**

Sol: Removal of electron from Mg^+ (requires IE_2) takes place from 3s-subshell, whereas the removal of electron from Mg^{2+} occurs from 2p-subshell and thus energy shell change during removal of electron is responsible from jump in IE values.

Q42 **Why do alkaline earth metals not form monovalent ions?**

Sol: The alkaline earth metal ions possess smaller size and thus have high heat of hydration which is responsible for the formation of divalent nature. Also, divalent cations of alkaline metals attain inert gas configuration.

Q43 **The atomic radius of Mg is smaller than sodium. Why?**

Sol: An increase in atomic number along the period results in an increase in effective nuclear charge and thus valence shells in magnesium atom are pulled more effectively towards nucleus resulting to smaller size of magnesium atom.

Q44 **KOH is a stronger base than $Ba(OH)_2$, Explain.**

Sol: The greater ionic radius and lower ionisation energy of potassium as compared to Ba results in stronger basic nature of KOH.

Q45 **$BeCl_2$ is soluble in ether but $BaCl_2$ is insoluble in ether. Explain.**

Sol: $BeCl_2$ being non-polar covalent molecule and thus soluble in non-polar or weak polar solvents. $BaCl_2$ is ionic and soluble in water.

Q46 **$MgCO_3$ is insoluble in water, why?**

Sol: Lattice energy of $MgCO_3$ is more than its hydration energy.

Q47 **Explain why $Be(OH)_2$ is soluble in NaOH but $Ba(OH)_2$ is insoluble in NaOH?**

Sol: $Be(OH)_2$ is amphoteric whereas NaOH is alkaline.

Q48 **$Be(OH)_2$ is amphoteric but $Mg(OH)_2$ is basic. Explain.**

Sol: Be has smaller size, so it is less ionized to release OH^- due to more covalent character.

Q49 **Beryllium halides fume in air, why?**

Sol: Due to formation of halogen acids on hydrolysis.

$$BeCl_2 + 2H_2O \rightarrow Be(OH)_2 + 2HCl\uparrow$$

Q50 **BeO and MgO have a high melting point, why?**

Sol: The lattice energy of BeO and MgO is appreciably higher due to their small size.

Q51 **Alkaline earth metal salts are less soluble than the corresponding alkali metal salts.**

Sol: Due to high charge-size ratio, lattice energy of alkaline earth metal salts are more than corresponding reducing gent and reduces SO_2 to S, using its oxygen for burning.

$$2Mg + SO_2 \rightarrow 2MgO + S$$

Q52 **Sodium and potassium carbonates are soluble in water but magnesium and calcium carbonates are insoluble in water. Why ?**

Sol: Lattice energy of Mg and Ca carbonates is much higher than sodium and potassium carbonate.

Q53 **Why is magnesium harder and denser than sodium ?**

Sol: Due to greater nuclear charge and smaller size, it packs more tightly in solid lattice.

Q54 **Why are salts of alkaline earth metal diamagnetic in nature?**

Sol: Alkaline earth metal ions have no unpaired electron and thus are diamagnetic in nature.

Q55 **Alkaline earth metals have higher m.pt. than alkali metals. Why?**

Sol: Alkaline earth metals possess two valence electrons and are much strongly bonded in the solid state than the alkali metals which results in higher m.pt.

Q56 **Alkaline earth metals are weaker reducing agent than alkali metals. Why?**

Sol: Due to high sublimation energy and high ionization energy, alkaline earth metals possess lower oxidation potential than alkali metals and thus they are less electropositive and weaker reducing agent than alkali metals.

Q57 **IE_1 of alkaline earth metals is very low as compared to their IE_2. Why?**

Sol: The much higher values of IE_2 for alkaline earth metals (just double to IE_1) are due to high heat of hydration of their cation in solution and high lattice energy in solid state.

Q58 **Be and Mg do not give flame colour whereas other alkaline earth metals do so. Why?**

Sol: Be and Mg atoms being smaller in size, bind their ns electrons more strongly i.e., their IE values are high. Hence, these possess high excitation energy and are not excited by the energy of the flame to higher energy level and thus de-excitation does not take. As a result no colour is produced in flame.

SOME IMPORTANT COMPOUNDS OF CALCIUM (Ca)

Important compounds of calcium(Ca) are calcium hydroxide ,calcium oxide, calcium sulphate, calcium carbonate, cement. These are industrially important compounds. The huge scale formulation of these compounds and their utilizes are described below.

Calcium Oxide or (Quick Lime / CaO)

CaCO3 on heating in rotary kiln at 1070-1270 K gives Calcium oxide. This method is used commercially for preparation of CaO

Calcium Oxide is made on a commercial scale by heating limestone ($CaCO_3$) in a rotary kiln at 1070-1270 K.

$$CaCO_3 \xrightarrow{\text{heat}} CaO + CO_2$$

CO_2 is taken out as soon as it is produced to enable the reaction to move in forward direction.

It is white Amorphous solid with a melting point 2870K. It soaks of moisture and CO_2 from atmosphere.

$$CaO + H_2O \rightarrow Ca(OH)_2$$
$$CaO + CO_2 \rightarrow CaCO_3$$

The addition of limited amount of water(H_2O) break into the lump of lime. This procedure is known as slaking of lime. Quick lime slaked with soda provides solid sodalime. Being a basic oxide, it react with acidic oxides at high temperature.

$$CaO + SiO_2 \rightarrow CaSiO_3$$
$$6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$$

Uses

It is an important primary material for manufacturing cement and is the cheapest form of alkali.

It(CaO) is used in the manufacturing of sodium carbonate from caustic soda.

Calcium Hydroxide (Slaked lime / $Ca(OH)_2$)

Calcium hydroxide($Ca(OH)_2$) is made by adding water to quick lime, CaO.

Calcium hydroxide($Ca(OH)_2$) is a white amorphous powder. It is sparingly soluble in water. The aqueous solution is called lime water and a suspension of slaked lime in water is called milk of lime.

When carbon dioxide is thrown through lime water it turns milky due to the formation of calcium carbonate($CaCO_3$).

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

On passing excess of carbon dioxide(CO_2), the precipitate dissolves to form calcium hydrogencarbonate($Ca(HCO_3)_2$).

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$$

Milk of lime react with chlorine to form hypochlorite, a component of bleaching powder($Ca(OCl)_2$).

$$2Ca(OH)_2 + 2Cl_2 \rightarrow Ca(OCl)_2 + CaCl_2 + 2H_2O$$

Utilizes:

(i) It is utilized in the preparation of mortar, a building material.

(ii) It is utilized in white wash due to its disinfectant nature.

(iii) It is utilized in glass making, in tanning industry, for the formulation of bleaching powder($Ca(OCl)_2$) and for purification of sugar.

Calcium Carbonate($CaCO_3$)

Limestone, chalk, marble are some examples where $CaCO_3$ is found in nature. It is white fluffy powder almost insoluble in water. It can be made by passing carbon dioxide through slaked lime or by the addition of sodium carbonate to calcium chloride.

By passing CO_2 through slaked lime or reaction of sodium carbonate with $CaCl_2$, $CaCO_3$ is formed.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaCl$$

On heating at 1200 K, it decomposes to evolve carbon dioxide.

$$CaCO_3 \xrightarrow{1200K} CaO + CO_2$$

On reaction with dilute acids, it evolves CO2.

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

$$CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$$

Uses of calcium carbonate:

In manufacturing of marble and in the manufacture of quick lime it is one important material. It is also used as flux in metallurgy for removal of acidic impurities from ore. In manufacturing of high-quality paper, specially precipitated $CaCO_3$ is extensively. It is also used as an antacid, mild abrasive in tooth paste, a component of chewing gum, and a filler in cosmetics.

Calcium Sulphate (Plaster of Paris / $CaSO_4 \cdot ½ H_2O$)

A hemihydrate of $CaSO_4$ is called plaster of Paris. When gypsum, $CaSO_4 \cdot 2H_2O$, is heated to 393 K POP is formed.

$$2(CaSO_4.2H_2O) \rightarrow 2(CaSO_4).½H_2O + 3H_2O$$

Above 393 K, water of crystallization evaporates out and anhydrous calcium sulphate is produced known as dead burnt plaster.

It has extraordinary property of setting with adequate amount of water making tough solid in 5 to 10 min.

Uses of POP:

In building industry for plaster. It is used for immobalising the affected part for fractured bones. The largest usage of Plaster of Paris is in the building industry as well as plasters, dentistry. It is also used as mold for ornaments.

Cement:

Cement or Portland cement, an important building material, was first introduced in England. Combining a material rich in lime, calcium oxide with materials like clay containing SiO_2, oxides of aluminium, iron and magnesium.

The average configuration of Portland cement is :

SiO_2, 20-25%; CaO, 50-60%; Al_2O_3, 5-10%; MgO, 2-3%; Fe_2O_3, 1-2% and SO_3, 1-2%. The ratio of silica (SiO_2) to alumina (Al_2O_3) should be among 2.5 and 4 for a great quality cement

- The amount of CaO to the sum of all of the oxides (SiO_2, Al_2O_3,Fe_2O_3) should around twice.
- The basic materials for the manufacturing of cement are limestone and clay. When clay and lime are strongly heated at once they fuse and react to form 'cement clinker'. This clinker is mixed with 2- 3% by weight of gypsum ($CaSO_4 \cdot 2H_2O$) to form cement. Thus important components present in Portland cement are tricalcium silicate (Ca_3SiO_5) 51% and dicalcium silicate (Ca_2SiO_4) 26%, tricalcium aluminate ($Ca_3Al_2O_6$) 11%.

Setting of Cement:

When it mixed with water, the setting of cement takes place to give a tough mass. This is expected to the hydration of the molecules of the constituents and their rearrangement. The purpose of combining gypsum is only to slow down the process of setting of the cement so that it gets suitably hardened.

Utilizes:

Cement has come to be a commodity of national essential for any country after that to iron and steel. It is utilized in concrete and reinforced concrete, in plastering and in the construction of dams ,bridges and buildings.

BIOLOGICAL SIGNIFICANCE OF MAGNESIUM AND CALCIUM

An adult body includes about 25 gm of Mg and 1200 gm of Ca compared with only 5 gm of iron and 0.06 gm of copper.

The daily necessity in the human body has been estimated to be 200mg. – 300 mg. All enzymes that use ATP in phosphate transfer involve magnesium as the cofactor.

The most important pigment for the absorption of light in plants is chlorophyll which includes magnesium(Mg). About 99% of body calcium is existing in teeth and bones.

The calcium concentration in plasma is measured at about 100 mgL^{-1}. It is kept by two hormones: parathyroid and calcitonin hormone.

Q59 **Describe the importance of the following :**
(a) Limestone, (b) Cement, (c) Plaster of Paris.

Sol: (a) In preparation of quick lime, as building material and road aggregate.
(b) In plastering and preparation of building dams, bridges, RCC.
(c) In surgical bandages, castings, molding and dentistry.

Q60 **Mention the main constituents or Portland cement.**

Sol: Dicalcium silicate (Ca_2SiO_4), tricalcium silicate (Ca_3SiO_3) and tricalcium aluminate ($Ca_3Al_2O_6$).

Q61 **What is milk of lime and for what purposes it is used?**

Sol: A solution-suspension of $Ca(OH)_2$ in water, used for white washing.

Q62 **What is precipitated chalk?**

Sol: The precipitate of $CaCO_3$ obtained during passage of CO_2 through lime water.

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 \downarrow + H_2O$$

Q63 **What is milk of magnesia and for what purposes it is used?**

Sol: A suspension of $Mg(OH)_2$ in water, used as antacid to remove acidity.

Q64 **Anhydrous calcium sulphate cannot be used as Plaster of Pairs. Why?**

Sol: ON moistening with water, anhydrous $CaSO_4$ does not set like Plaster of Paris.

Q65 **NaCl is added to anhydrous $MgCl_2$ (fused) before electrolysis. Why?**

Sol: The addition of NaCl to anhydrous $MgCl_2$ during its electrolysis is done to lower the fusion point of $MgCl_2$ as well as to increase the conducting power of fused mixture.

Q66 **Name an element which is invariable bivalent and whose oxide is soluble in excess of NaOH and its dipositive ion has a noble gas core.**

Sol: Be : $1s^2, 2s^2$

$$BeO + 2NaOH \longrightarrow \underset{\text{Soluble}}{NaBeO_2} + H_2O$$

$Be^{2+} = 1s^2$ [Noble gas He core]

Q67 **Gypsum is added to powdered clinkers to get cement. Why?**

Sol: Addition of gypsum to clinker decreases its setting time and thus cement hardens readily on mixing with water.

Q68 **MgO is used for lining of steel making furnace. Why?**

Sol: MgO is acts as basic flux and facilitates the removal of acidic impurities of Si, P and S from steel through slag formation.

(i) Magnesium Peroxide (MgO_2) and Calcium Peroxide (CaO_2) are obtained by passing H_2O_2 in a suspension of $Mg(OH)_2$ and $Ca(OH)_2$.

(ii) MgO_2 is used as an antiseptic in tooth paste and as a bleaching agent.

(iii) **Preperation of NaOH :** Caustication of Na_2CO_3 (Gossage's method):

$Na_2CO_3 + Ca(OH)_2 \rightarrow 2NaOH + CaCO_3\downarrow$

(suspension)

Since the K_{sp} $(CaCO_3)$ < K_{sp} $(Ca(OH)_2)$, the reaction shifts towards right.

(iv) As a reagent KOH is less frequently used but in absorption of CO_2, KOH is preferably used compared to NaOH. Because $KHCO_3$ formed is soluble whereas $NaHCO_3$ is sparingly soluble and may therefore choke the tubes of apparatus used.

(v) Calcium hydroxide is used as a mortar.

[Mortar is a mixture of slaked lime (1 Part) and sand (3 Parts) made into paste with water.]

(vi) NaCl is used to prepare freezing mixture in laboratory [Ice-common salt mixture is called freezing mixture and temperature goes down to –23°C.]

(vii) On heating $MgCl_2 \cdot 6H_2O$ undergoes hydrolysis as follows:

$$MgCl_2 \cdot 6H_2O \xrightarrow{\Delta} Mg(OH)Cl + HCl + 5H_2O$$

$$Mg(OH)Cl \rightarrow MgO + HCl$$

** Hence, Anh. $MgCl_2$ cannot be prepared by heating this hydrate.

** Because of the formation of HCl , sea water cannot be used in marine boilers which corrodes the iron body.

(viii) Anhydrous $MgCl_2$ can be prepared by heating a double salt like $MgCl_2 . NH_4Cl . 6H_2O$ as follows:

$$MgCl_2 . NH_4Cl . 6H_2O \xrightarrow[\Delta]{-H_2O} MgCl_2 . NH_4Cl \xrightarrow[\Delta]{strong} MgCl_2 + NH_3 + HCl$$

(ix) Sorel Cement is a mixture of MgO and $MgCl_2$ (paste like) which set to hard mass on standing, this is used in dental filling, flooring etc.

(x) Anh. $CaCl_2$ is used in drying gases and organic compounds but not NH_3 or alcohol due to the formation of $CaCl_2 . 8NH_3$ and $CaCl_2 . 4C_2H_5OH$.

(xi) One interesting feature of the solubility of glauber's salt is; when crystallised at below 32.4°C, then $Na_2SO_4 . 10H_2O$ is obtained but above 32.4°C, Na_2SO_4 (anh.) comes out.

(xii) Leblanc Process (Preparation of Na_2CO_3) :

$$NaCl + H_2SO_4 \text{ (conc.)} \xrightarrow{\text{mild heating}} NaHSO_4 + HCl$$

$$NaCl + NaHSO_4 \xrightarrow[\text{heated}]{\text{strongly}} Na_2SO_4 + HCl$$

(Salt Cake)

$$Na_2SO_4 + 4C \rightarrow Na_2S + 4CO\uparrow$$

$$Na_2S + CaCO_3 \rightarrow Na_2CO_3 + CaS$$